

AI Engineer + Full Stack Interview Roadmap

Roadmap

Phase 1: Master LLM APIs

1. Basic LLM API Calls

- Responses API and Chat Completions
- temperature, top_p, max_tokens
- system/user/assistant roles
- reasoning vs chat models

Interview: Responses API vs Chat Completions, context length, temperature vs top_p.

2. Streaming

- stream=True
- delta events
- stop conditions

Interview: Why streaming? SSE vs WebSockets? Streaming to React/Vue.

3. Conversation Memory

- Maintain messages list manually
- Trimming
- Summarization
- Token counting

Interview: Handling long conversations.

4. Tool / Function Calling

- Single tool
- Multiple tools
- Parallel tools
- Auto vs forced tool choice

Interview: Why tools instead of letting LLM access APIs?

5. Streaming + Tool Calling

- Stream text
- Detect tool calls
- Execute tools
- Resume streaming

Interview: Explain ChatGPT architecture.

Phase 2: Core AI Concepts

6. Structured Outputs (Pydantic/JSON)

7. JSON Mode

8. Embeddings

- Cosine similarity
- Dot product
- Normalization

Interview: Embeddings vs LLM.

9. Vector Databases

- Learn one well (e.g. ChromaDB)
- Retrieval using nearest neighbors

Interview: Why not MySQL?

10. Retrieval Augmented Generation (RAG)

Pipeline:

PDF -> Chunk -> Embedding -> Vector DB -> Retrieve -> Prompt -> LLM

Interview: Fine-tuning vs RAG.

Phase 3: Agent Engineering

11. Multi-tool Agents

Weather, Calculator, Search, SQL, Email, Filesystem.

Interview: How does the LLM decide which tool to call?

12. Agent Loop

LLM -> Tool? -> Execute -> Continue

Interview: Explain ReAct.

13. Multi-Agent Systems

Planner, Researcher, Executor, Reviewer.

Interview: When to use multiple agents?

14. Memory

Short-term, long-term, semantic memory, summaries.

Interview: Why not send the whole conversation?

Phase 4: Production AI

15. Prompt Engineering

- System prompts
- Few-shot
- Chain-of-thought (concept)
- Self-consistency
- Role prompting

Interview: Reducing hallucinations.

16. Token Management

Input/output tokens, cost, latency, caching.

17. Rate Limits

Retries, exponential backoff, timeouts, circuit breakers.

18. Guardrails

Moderation, prompt injection, tool/output validation.

Interview: Handling 'Ignore previous instructions'.

19. Evaluation

BLEU, ROUGE, BERTScore (high level), LLM-as-a-Judge, golden datasets.

Phase 5: Frameworks

- LiteLLM
- LangGraph
- PydanticAI
- LlamaIndex
- OpenAI Agents SDK

Understand what they abstract before using them.

Phase 6: AI System Design

Design ChatGPT:

Frontend -> Backend -> Redis -> Conversation Store -> Tool Manager -> Vector DB -> LLM
Gateway -> Provider.

Design GitHub Copilot:

Editor -> Retriever -> Prompt Builder -> LLM -> Streaming.

Design AI Resume Analyzer:

PDF Upload -> OCR -> Chunk -> Embeddings -> Retriever -> LLM -> Structured JSON.

Suggested Weekly Plan

Week 1: APIs, streaming, conversation history, tool calling, structured outputs.

Week 2: Embeddings, vector DBs, RAG from scratch.

Week 3: Multi-tool agents, memory, retries, streaming.

Week 4: System design, prompt engineering, caching, rate limits, guardrails.

Week 5: LangGraph, LiteLLM, PydanticAI.

Week 6+: Production-grade portfolio projects.

Portfolio Projects

1. Chat with PDFs (RAG)
2. SQL Agent
3. GitHub Repository Assistant

4. Multi-tool AI Assistant

5. AI Code Review Bot

Learning Strategy

For every topic:

- Understand the concept.
- Implement from scratch (without LangChain).
- Learn common interview questions.
- Learn production considerations.
- Finally learn how frameworks simplify it.